

ETF Rotation Strategy Logic & Transaction Cost Accounting

Technical Specification Document

Document Information

- **Version:** 1.0
 - **Date:** July 2025
 - **Purpose:** Business validation of strategy logic and cost accounting
 - **Compliance:** Indian tax laws and market regulations
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Executive Summary

This document details the implementation of an ETF rotation strategy with comprehensive Indian market cost accounting, including capital gains tax compliance. The strategy operates in two phases: accumulation and churning, with all transactions subject to realistic market costs and tax implications.

1. Strategy Overview

1.1 Core Concept

The ETF rotation strategy is a momentum-based investment approach that:

- **Accumulates ETF positions** during an initial phase
- **Rotates capital** from high-performing to low-performing ETFs during churning phase
- **Uses 52-week high/low momentum** as the primary selection criterion

1.2 Strategy Phases

Phase 1: Accumulation (Weeks 1-52, configurable)

- Weekly investment of fixed capital amount
- Purchase ETF closest to 52-week low
- Build diversified portfolio foundation

Phase 2: Churning (Week 53 onwards)

- Sell ETFs closest to 52-week high
 - Buy ETFs closest to 52-week low
 - Maintain momentum-based rotation
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2. Detailed Strategy Logic

2.1 Signal Generation Process

2.1.1 Timing Framework

Signal Date: Friday of current week (or nearest trading day)

Execution Date: Monday of following week (or nearest trading day)

2.1.2 52-Week Momentum Calculation

For each ETF at signal date:

- **52-week high:** Maximum closing price in trailing 252 trading days
- **52-week low:** Minimum closing price in trailing 252 trading days
- **Distance from low:** $(\text{Current Price} - 52\text{w Low}) / 52\text{w Low} \times 100$
- **Distance from high:** $(52\text{w High} - \text{Current Price}) / 52\text{w High} \times 100$

2.2 Accumulation Phase Logic

2.2.1 Selection Criteria

python

Pseudo-code for accumulation

target_etf = ETF_with_smallest_distance_from_52w_low

capital_allocation = weekly_capital_amount

execution_price = monday_open_price

2.2.2 Purchase Calculation

1. Calculate transaction costs on intended purchase amount
2. Determine maximum units purchasable with available cash
3. Execute purchase at Monday opening price
4. Record purchase in FIFO tracking system

2.3 Churning Phase Logic

2.3.1 Capital Raising Process

python

Pseudo-code for churning

target_capital = weekly_capital_amount

etfs_ranked_by_distance_from_high = sort_descending(distance_from_52w_high)

for each etf in etfs_ranked_by_distance_from_high:

 if total_raised >= target_capital:

 break

```

sell_units = min(available_units, required_units_for_remaining_capital)

execute_sell_with_costs_and_taxes()

```

2.3.2 Reallocation Process

```
python
```

```
# After raising capital
```

```
target_etf = ETF_with_smallest_distance_from_52w_low
```

```
available_capital = total_raised_from_sells
```

```
execute_purchase_with_costs()
```

3. Transaction Cost Accounting

3.1 Indian Market Cost Structure

3.1.1 Cost Components (Per Transaction)

Cost Type	Rate	Applied On	When
Brokerage	User-defined %	Transaction Value	Buy & Sell
STT (Securities Transaction Tax)	0.001%	Transaction Value	Sell Only
Stamp Duty	0.005%	Transaction Value	Buy Only
Exchange Charges	0.00345%	Transaction Value	Buy & Sell
SEBI Charges	0.0001%	Transaction Value	Buy & Sell
GST	18%	Brokerage Amount	Buy & Sell

3.1.2 Cost Calculation Formula

Total Transaction Costs = Brokerage + STT + Stamp Duty + Exchange Charges + SEBI Charges + GST

Where:

- Brokerage = Transaction Value × (Brokerage % / 100)
- STT = Transaction Value × 0.001% (sell only)
- Stamp Duty = Transaction Value × 0.005% (buy only)
- Exchange Charges = Transaction Value × 0.00345%
- SEBI Charges = Transaction Value × 0.0001%
- GST = Brokerage × 18%

3.1.3 Net Amount Calculation

For Purchases: Net Amount = Transaction Value + Total Costs

For Sales: Net Amount = Transaction Value - Total Costs

3.2 Capital Gains Tax Implementation

3.2.1 FIFO (First-In-First-Out) Logic

The system maintains detailed purchase history for each ETF:

python

Purchase Record Structure:

```
{
    'date': execution_date,
    'units': units_purchased,
    'price': purchase_price,
    'remaining_units': units_available_for_sale
}
```

3.2.2 Capital Gains Calculation Process

1. **Identify Oldest Purchases:** Sort purchase records by date (ascending)
2. **Allocate Sale Units:** Match sale units against oldest available purchase units
3. **Calculate Profit/Loss:** For each matched batch:
 4. Cost Basis = Units Sold × Original Purchase Price
 5. Sale Value = Units Sold × Current Sale Price

Profit/Loss = Sale Value - Cost Basis

6. **Apply Tax Rate:** Capital Gains Tax = Max(0, Total Profit × 12.5%)

3.2.3 FIFO Example

Purchase History for ETF_A:

Batch 1: 100 units @ ₹50 (₹5,000) - Date: Jan 1

Batch 2: 150 units @ ₹55 (₹8,250) - Date: Feb 1

Batch 3: 200 units @ ₹60 (₹12,000) - Date: Mar 1

Sale Transaction: 180 units @ ₹65

FIFO Allocation:

- 100 units from Batch 1: Profit = (65-50) × 100 = ₹1,500

- 80 units from Batch 2: Profit = (65-55) × 80 = ₹800

Total Profit = ₹2,300

Capital Gains Tax = ₹2,300 × 12.5% = ₹287.50

3.3 Cash Flow Impact

For Each Transaction:

Cash Impact = Net Transaction Amount ± Capital Gains Tax

Portfolio NAV = Cash + Sum(Holdings × Current Prices)

4. Performance Measurement

4.1 Return Calculations

4.1.1 Total Return

Total Return = (Final NAV - Total Investment) / Total Investment × 100

4.1.2 CAGR (Compound Annual Growth Rate)

CAGR = ((Final NAV / Total Investment)^(1/Years) - 1) × 100

4.1.3 XIRR (Extended Internal Rate of Return)

Uses Newton-Raphson method considering actual cash flow timing:

- **Outflows:** Weekly investments during accumulation
- **Inflow:** Final portfolio value
- **Calculation:** Solves for rate where NPV of all cash flows = 0

4.2 Risk Metrics

4.2.1 Volatility

Annualized Volatility = Weekly Return StdDev × √52 × 100

4.2.2 Maximum Drawdown

Drawdown = (NAV - Peak NAV) / Peak NAV × 100

Max Drawdown = Minimum of all drawdown values

4.2.3 Sharpe Ratio

Sharpe Ratio = (CAGR - Risk Free Rate) / Volatility

4.2.4 Treynor Ratio

Treynor Ratio = (CAGR - Risk Free Rate) / Beta

5. Benchmark Comparison

5.1 Nifty50 Buy-and-Hold Strategy

5.1.1 Implementation

- **Investment:** Lump sum equal to total ETF strategy investment
- **ETF Used:** NIFTYBEES (Nifty50 ETF)
- **Timing:** Single purchase at strategy start date
- **Costs:** Same transaction cost structure applied

5.1.2 Fair Comparison Framework

ETF Strategy Total Investment = Accumulation Weeks × Weekly Capital

Nifty50 Strategy Investment = Same total amount invested upfront

Both strategies include identical cost structures

5.2 Performance Metrics Comparison

Both strategies calculated using identical methodologies:

- Returns, CAGR, XIRR
 - Risk metrics (volatility, drawdown, ratios)
 - Cost impact analysis
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6. Data Requirements

6.1 ETF Universe Selection

- **Minimum Data History:** Strategy automatically calculates common date range
- **Data Quality:** OHLCV data with corporate action adjustments
- **52-Week Calculation:** Requires minimum 252 trading days for accurate momentum signals

6.2 Trading Calendar

- **Signal Day:** Friday (fallback: Thursday if Friday is holiday)
 - **Execution Day:** Monday (fallback: Tuesday if Monday is holiday)
 - **Holiday Handling:** Automatic detection using available trading dates
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7. Cost Impact Analysis

7.1 Transaction Cost Tracking

Every transaction generates detailed cost breakdown:

- Individual cost components
- Cumulative cost tracking

- Cost as percentage of transaction volume

7.2 Capital Gains Tax Tracking

- FIFO-based tax calculation for each sale
- Cumulative tax burden tracking
- Integration with total cost analysis

7.3 Performance Impact Assessment

Return Impact = Returns Without Costs - Returns With Costs

Cost Efficiency = Total Costs / Total Transaction Volume × 100

8. Risk Disclosures

8.1 Strategy Risks

- **Momentum Risk:** Strategy assumes momentum persistence
- **Transaction Cost Drag:** High turnover may erode returns
- **Tax Efficiency:** Frequent trading triggers capital gains tax

8.2 Implementation Assumptions

- **Liquidity:** Assumes sufficient ETF liquidity for trades
- **Execution:** Monday opening prices used for all trades
- **Costs:** Based on current Indian market cost structure

8.3 Backtesting Limitations

- **Survivorship Bias:** Only includes ETFs with complete data
 - **Market Impact:** Assumes no price impact from trades
 - **Regulatory Changes:** Cost structure based on current regulations
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9. Validation Checklist

9.1 Strategy Logic Validation

- 52-week momentum calculation accuracy
- FIFO purchase tracking implementation
- Phase transition logic (accumulation to churning)
- ETF selection criteria in each phase

9.2 Cost Accounting Validation

- Indian market cost rates and application

- Capital gains tax calculation methodology
- FIFO cost basis determination
- Net cash flow impact calculation

9.3 Performance Calculation Validation

- XIRR calculation with actual cash flow timing
 - Risk metric calculations (Sharpe, Treynor, drawdown)
 - Benchmark comparison methodology
 - Cost impact analysis accuracy
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10. Technical Implementation Notes

10.1 Database Structure

- ETF OHLCV data with date indexing
- Metadata table with ETF availability periods
- Support for multiple data sources

10.2 Calculation Engine

- Weekly iteration framework
- Date handling for trading calendars
- Portfolio tracking with holdings and cash

10.3 Output Generation

- Comprehensive performance metrics
 - Transaction cost analysis
 - Comparative benchmarking
 - Export capabilities for further analysis
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Document Prepared For: Business User Validation

Technical Review Required: Strategy logic, cost calculations, tax compliance

Regulatory Compliance: Indian securities market regulations and tax laws
